CogniFarm: A Gamified AI Companion for Cognitive Health Monitoring in Seniors

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Presentation Outline

1. Background and Motivation
2. Literature Review and Gap Analysis
3. System Design
4. AI System
5. Study Design
6. Results
7. Conclusion

The Problem

55M+
People with dementia worldwide (WHO, 2025)

10M
New cases/year

10-15%
Annual conversion rate (Petersen & Negash, 2008)

The window for early intervention exists, but only if cognitive decline is detected early

Why is early detection hard?

- Clinic visits are infrequent. Cognitive changes between visits go unnoticed
- No accessible daily cognitive monitoring tools designed for seniors at home
- Existing digital tools lack AI personalisation and longitudinal trend tracking

What already exists and what's missing

FINGER Trial (Ngandu et al. , 2015): Evidence base

- 1260 participants (60-77 years old) - randomised controlled trial
- Multidomain: cognitive training + exercise + diet + vascular monitoring
- 25% better composite cognitive score vs control group

Existing tools

MMSE / MoCA: Clinical standard but clinic-only, 6–12 monthly

Lumosity: Preset difficulty tiers, no adaptive AI

BrainHQ: Fixed exercise protocols, no personalisation

Elevate: Rule-based personalisation

3.3% 30-day retention (Baumel et al., 2019)

Feature	BrainHQ/CogniFit	FINGER Trial	CogniFarm
RL-adaptive difficulty per session	✗ (rule-based/proprietary)	✗	✅
Automated trend decline alerts	✗	Partial (manual)	✅
RL-personalised reminders	Partial (fixed rules)	✗	✅
Mobile, daily at-home-use	✅	✗ (clinic)	✅

CogniFarm – What was built

Adaptive Difficulty AI with RL

Q-Learning adjusts challenge each session.
Targets 60–80% accuracy (cognitive flow zone).

Smart Reminder AI

Thompson Sampling learns best notification type + hour.
Personalised per player over time.

Cognitive Trend Detection

Monitors accuracy slope across 5-session windows.
Alerts after 8+ sessions of consistent decline.

Harvesting Memory Game

Sequence memory (Corsi Block Tapping Test)

- Watch crops light up in sequence
- Repeat sequence by tapping plots
- Difficulty = sequence length (3-7 items)

AI adjusts sequence length each round

Planting Seeds Game

Card matching (Paired Associates Learning)

- Cards revealed briefly (preview phase)
- Flip and match seed card pairs from memory
- Difficulty = pairs count

AI adjusts pairs count

Both games share the same Q table. Cross-game learning is intentional

Adaptive Difficulty AI

Q-Learning State Space

Performance: <40% Poor / 40–60% Fair / 60–80% Good / ≥80% Excellent
Response time: >10 s Slow / 5–10 s Normal / <5 s Fast
Difficulty: Level 1-5

4 x 3 x 5 = 60 states x 3 actions = 180 Q-Values

Actions: Decrease/ Maintain/ Increase Difficulty

ε-greedy: ε = 0.3 → decays to 0.02 over ~60 sessions
α = 0.2 γ = 0.9 Q-table saved across sessions

Reward Function:

60-80% Flow Zone	40-90% Acceptable	>90-% Too easy	<40% Too hard
+1.0	+0.5	-0.3	-0.5

Response time: <8s: +0.3, > 15s: -0.2

Q-Learning vs Rule-based

Scenario	Rule-based	Q-learning
91% accuracy, FAST response	Increase ↑	Increase ↑
91% accuracy, SLOW response	Increase ↑ (ignores response time)	Maintain → (fatigue state)
70% accuracy. FAST response	Maintain →	Maintain → (flow zone)
35% accuracy	Decrease ↓	Decrease ↓
Abandoned session	Log only (no penalty)	-1.5x(1-ratio) Penalty to Q-Table

State space: what each AI actually sees

Q-Learning: In-Game Difficulty AI

60 states × 3 actions = 180 Q-values

Accuracy (Performance)	< 40% Poor	40–60% Fair	60–80% Good	≥ 80% Excellent	
Avg Response Time	> 10 s Slow	5–10 s Normal	< 5 s Fast		
Current Difficulty	1 V.Easy	2 Easy	3 Medium	4 Hard	5 V.Hard

Actions: ↓ Decrease → Maintain ↑ Increase difficulty

Thompson Sampling: Reminder AI

192 states × 3 actions = 576 Beta posteriors

4 × 4 × 4 × 3 = 192 states

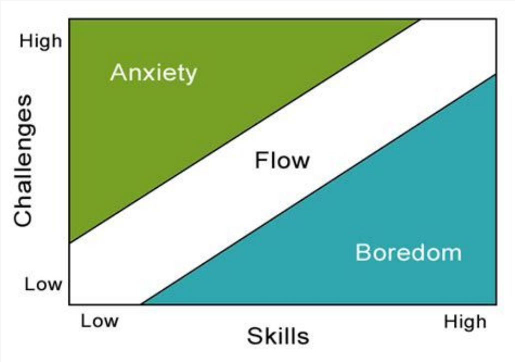
Time of Day	Morning 6–12h	Afternoon 12–17h	Evening 17–21h	Night 21–6h
Days Inactive	0 days (today)	1 day (yesterday)	2 days ago	3+ days ago
Last Response	None (first time)	Accepted ✅	Rejected ✗	Ignored –
Streak Status	No streak 0 days	Short 1–2 days	Long 3+ days	

0 Don't send	**1 Gentle reminder**	**2 Urgent reminder**

Beta(α,β) updated per state-action: player opens app → α+1 / ignores → β+1

Why Q-Learning?

Flow Theory (Csikszentmihalyi, 1990)



Flow Zone = 60-80% accuracy

Q-Learning: designed for flow zone

- ✅ Rewards 60–80% accuracy with +1.0
- ✅ Penalises >90% accuracy (−0.3)
- ✅ Tracks response time — fatigue ≠ mastery
- ✅ ε=0.3: explores before converging on policy

Rule-Based: no flow awareness

- ✗ Increases difficulty at ≥80% — already above optimal
- ✗ No reward penalty for accuracy >90%
- ✗ Ignores response time completely
- ✗ All 8 RB players hit difficulty ceiling (peak=5.0)

Study Preview

Q-Learning

56.8% in flow zone

Rule-Based

37.2% in flow zone

+19.6 pp difference
p=0.091 r=0.39

Two AI Companion Features

Reminder AI — Thompson Sampling

Why Thompson Sampling?

- ✅ No ε to tune
- ✅ Beta (1,1) honest about cold-start
- ✅ Proven in mHealth

TS 98.4% acceptance

RB 29% acceptance

Simulation on HeartSteps V1 — validates algorithm, not study duration

Two-layer Architecture

On-device: 192 states × 3 actions (don't / gentle / urgent) · Beta(α,β) per state-action

Cloud: 24-arm Beta per player → learns optimal send hour

Cognitive Trend Detection

Every session	Record accuracy, response time, difficulty
5-session window	OLS linear regression over last 5 sessions
Slope > +2%/session	→ IMPROVING ✅
Otherwise	→ STABLE
Slope < −15%/session + ≥8 sessions	→ DECLINING ⚠️ Alert triggered

Alert shown in-app:

"We've noticed a steady decline over several sessions. Consider speaking with a healthcare professional."

Not diagnostic wellness signal only

8-session gate prevents false alarms

Local only no data shared externally

Study Design

N = 18

participants

Group A — Rule-Based

8 players · 190 sessions

Fixed thresholds:
$\geq 80\% \rightarrow$ increase / $\leq 40\% \rightarrow$ decrease

Group B — Q-Learning

10 players · 160 sessions

Reward-based:
Flow zone = +1.0
Abandon penalty = $-1.5\times(1-\text{ratio})$

Timeline

Mar 24 — Study start

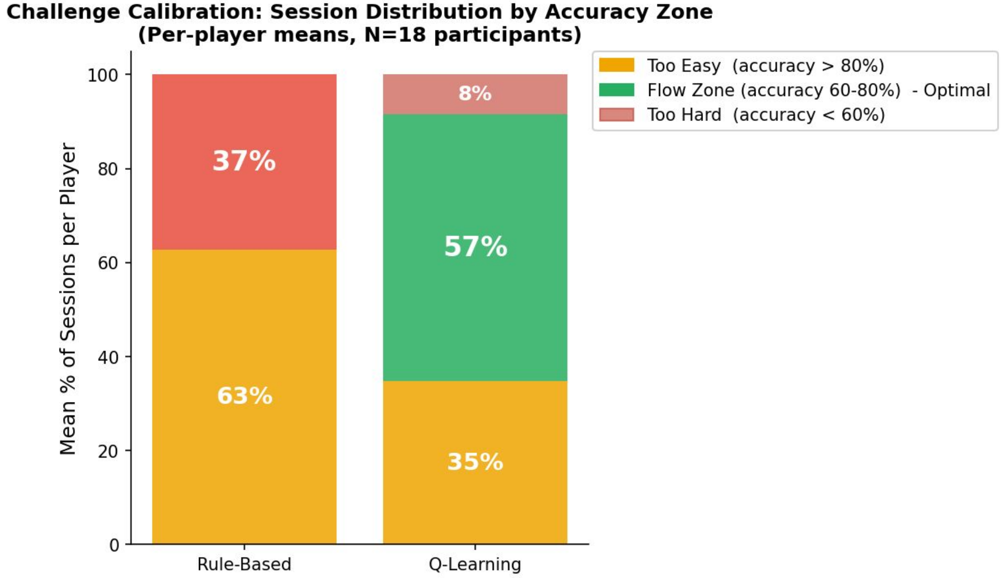
Active Data Collection

Apr 13 — Study end

Outcome Measures

- Flow zone % (60–80% acc.)
- Mean accuracy
- Difficulty progression
- Avg response time
- Reminder engagement

Results: Challenge Calibration



Flow Zone Sessions (per-player mean)

Q-Learning 56.8%
± 25.8% (n=10)

Rule-Based 37.2%
± 25.1% (n=8)

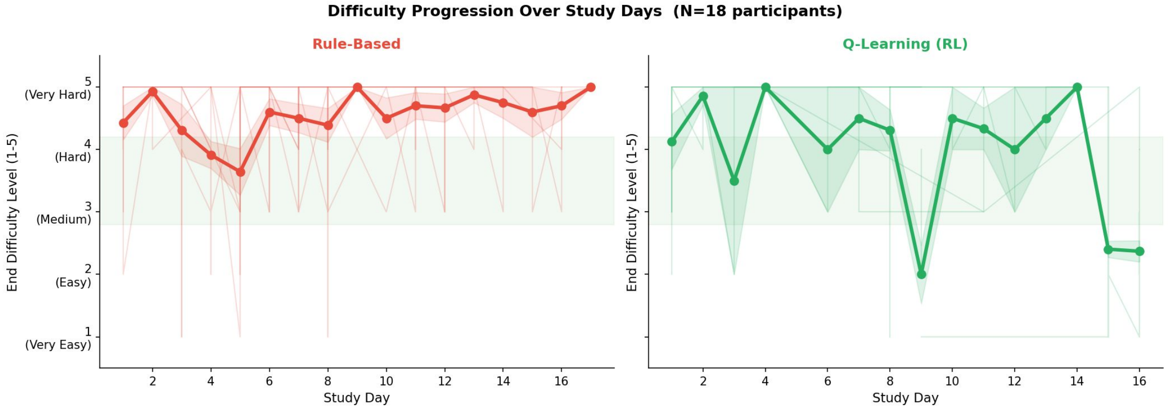
+19.6 percentage point difference

Mann-Whitney U (one-tailed)

p = 0.091 · r = 0.39 (medium effect)

N=18 underpowered. Directional trend is clear

Results: Difficulty Progression & Key Metrics



Key Metrics Summary

Accuracy

QL 78.5% vs RB 91.3%

QL in flow zone, RB overshooting

Difficulty Progression

QL +0.79 vs RB +0.48

QL adapted difficulty level faster

Response Time

QL 8.6 s vs RB 9.3 s

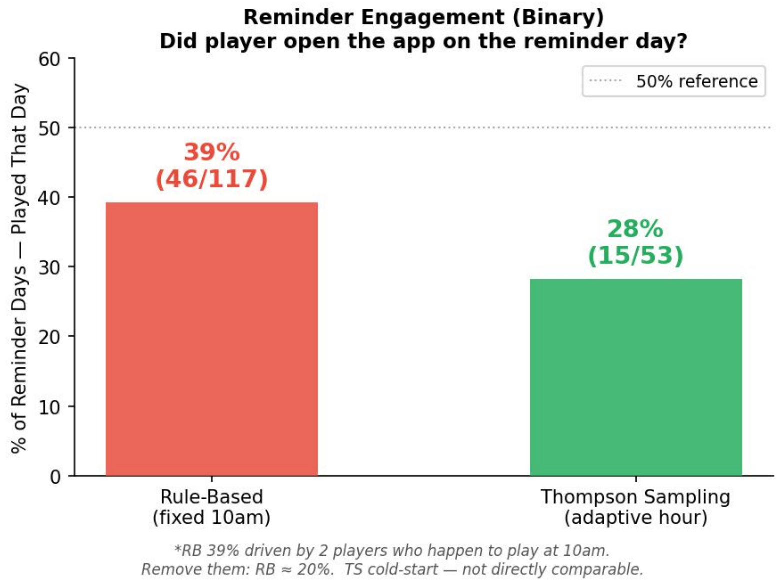
QL slightly more cognitively engaged

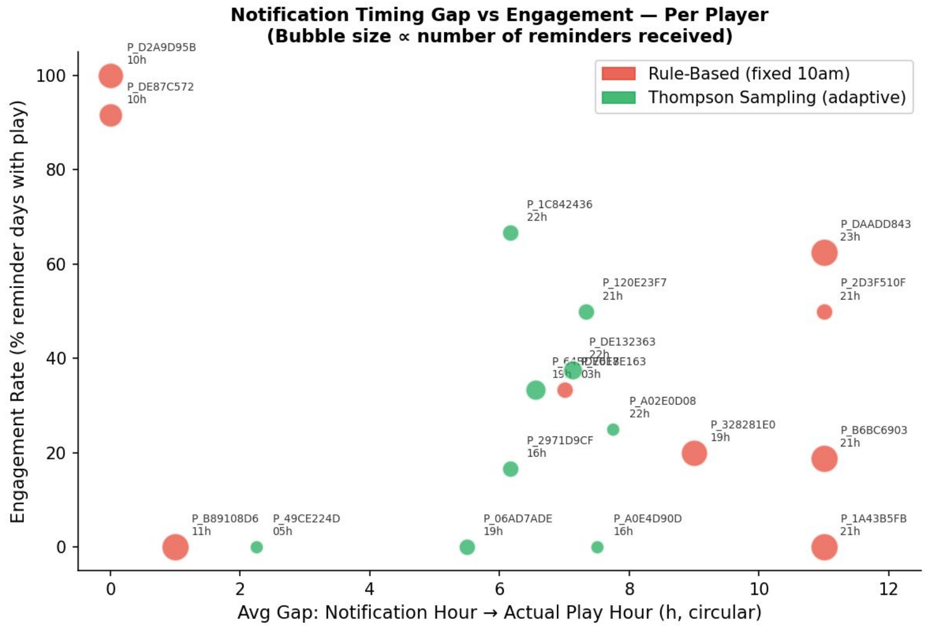
Peak Difficulty

QL 4.90 vs RB 5.00

RB hit ceiling, QL calibrated

Results: Reminder Engagement





RB 39.3% may not be a fair win

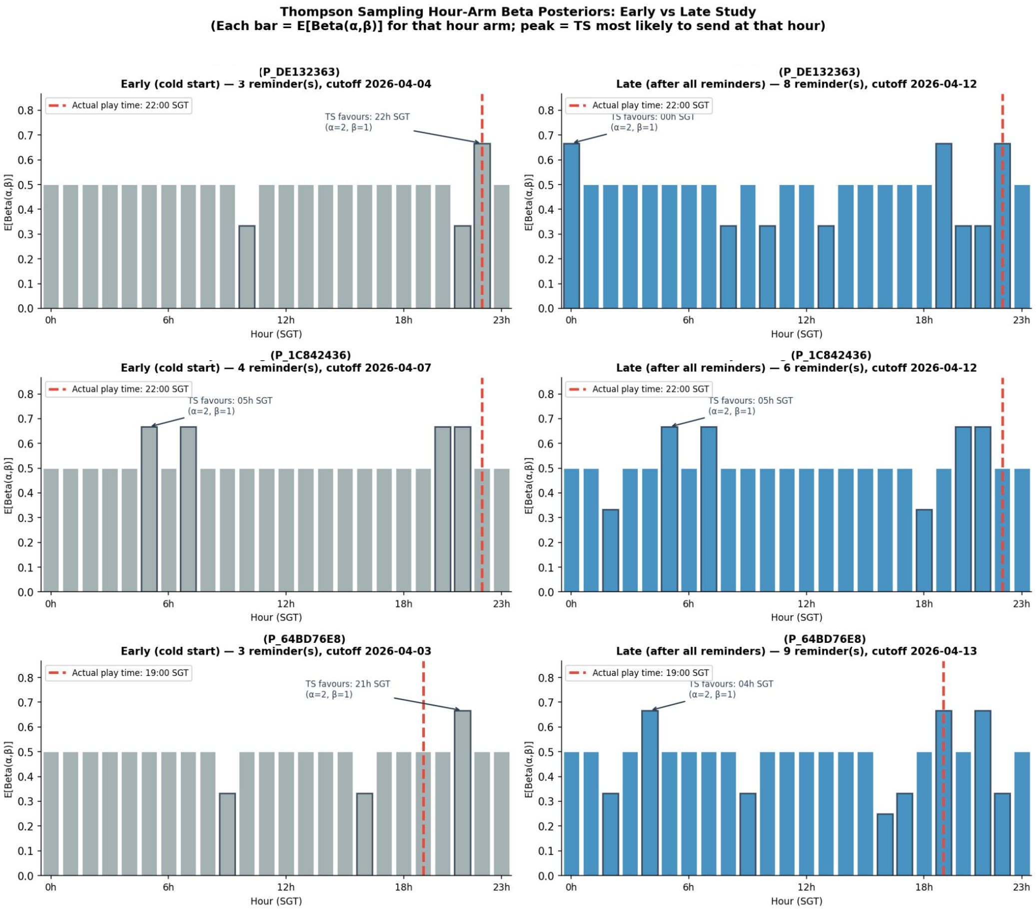
2 players coincidentally play at 10am = RB's fixed send time → 100% + 92% engagement.
If we remove them: remaining 7 RB players average ~20%.

TS 28.3%: cold-start expected

~6 reminders/player in 16 days — 24-arm posteriors need 30+ to converge.
TS reaches 98% acceptance in simulation (HeartSteps, Oralytics).
Algorithm is valid; study was too short.

TS Convergence – Real player data

Early (grey): flat posteriors, TS uncertain.
Late (blue): peaks emerging, TS identifying preferred hour. Needs 30+ reminders to fully converge.



Literature Validation — RL Algorithms on Real mHealth Data

Algorithm Acceptance Rate — HeartSteps V1 & Oralytics

Algorithm	HeartSteps V1	Oralytics Pilot	Average	Notes
Thompson Sampling	**98.4%**	**98.3%**	**98.4%**	Best overall ✅ Our reminder algo
Epsilon-Greedy	69.3%	62.2%	65.7%	Good general baseline
LinUCB	57.0%	62.2%	59.6%	Contextual bandit
Q-Learning	48.3%	61.3%	54.8%	Our difficulty AI method
Rule-Based	29.0%	62.9%	46.0%	Phase 1 baseline ← ours
Random	26.9%	61.3%	44.1%	Worst case floor

Takeaway: Thompson Sampling achieves near-perfect acceptance on both HeartSteps and Oralytics when given sufficient data. Our 21-day study is too short for convergence, the simulation validates the algorithm's long-run potential independently of study duration. Q-Learning outperforms Rule-Based on HeartSteps (48% vs 29%).

Limitations

Sample size (N=18)

- $p=0.091$ does not cross $\alpha=0.05$.
- Effect size $r=0.39$ is medium-large and clinically meaningful, but statistical power is insufficient.

Study duration (21 days)

- Thompson Sampling 24-arm posteriors need 30+ reminders per player to converge.
- Average ~6–12 reminders per player in this study means TS was still in cold-start.
- Reminder results are inconclusive but not negative.

Non-clinical population

- Recruited participants are family and friends, not seniors with cognitive concerns. Target demographic may show different engagement patterns.

Android only (study)

- iOS distribution required an Apple Developer Account and app store review – outside of project scope

Conclusion – Answered research questions

1

Challenge calibration

✅ **Supported**

- Q-Learning: 56.8% flow zone vs Rule-Based: 37.2%
- +19.6 pp difference · p=0.091 · r=0.39 (medium effect)
- Directional trend clear despite small N.

2

Reminder personalisation

Not definitive

- TS could not converge in 16 days (~6 reminders/player)
- RB's 39% advantage explained by 2 coincidentally time-matched players.
- TS achieves 98.4% acceptance in simulation.

3

Cognitive trend detection

✅ **Supported**

- 5-session OLS regression monitors accuracy + response time slopes.
- Alert fires after 8+ sessions of sustained decline.
- Not triggered: consistent with healthy participant pool.

Live Demo – CogniFarm on Android

1) Main menu + onboarding flow

2) Harvesting Memory Game: difficulty adapting across rounds

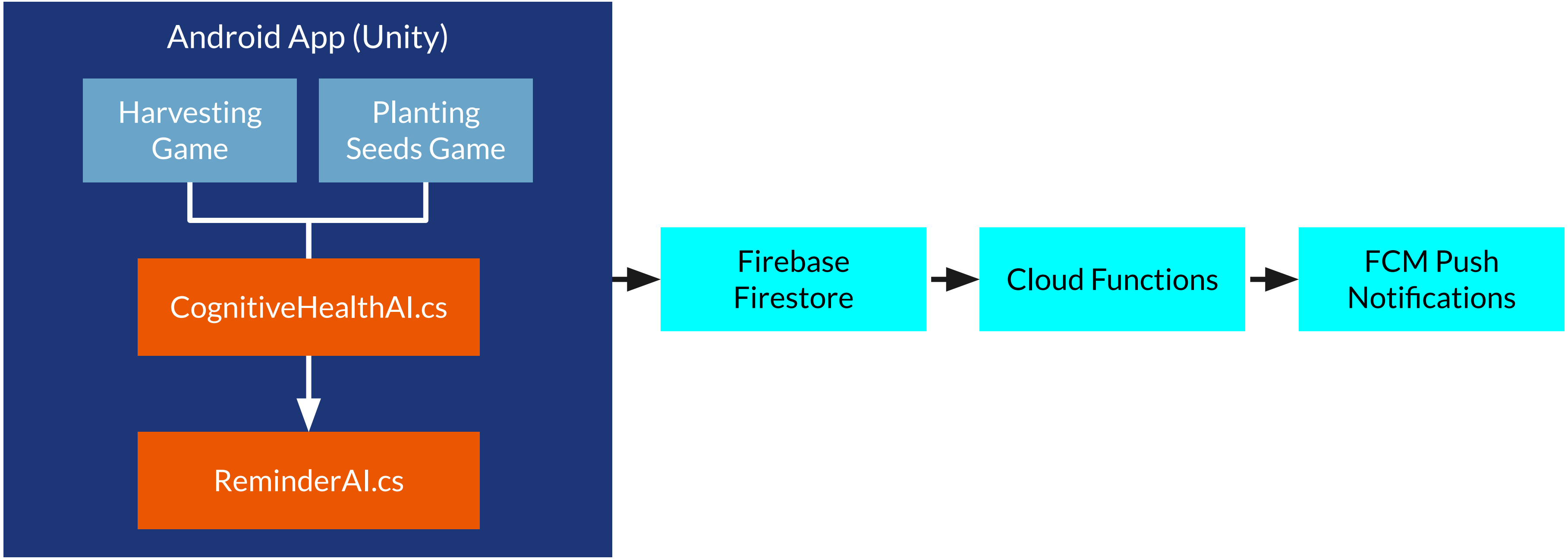
3) Planting Seeds Game: card matching + result screen

4) Result screen: trend report + AI status summary

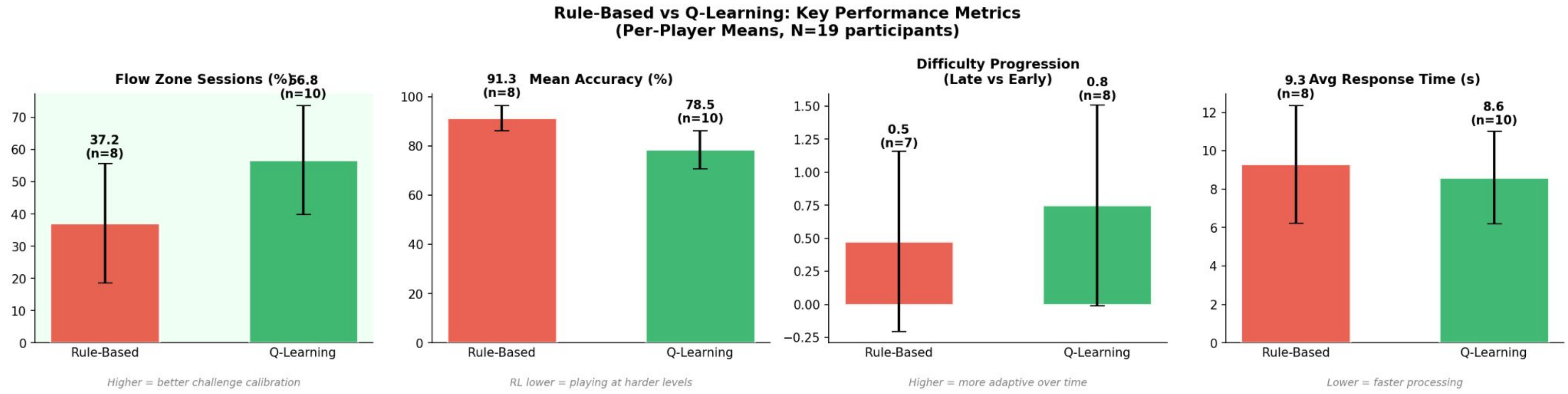
5) Firebase Firestore

Thank you!

System Architecture



Experiment 2: User study



CogniFarm – What was built

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their effectiveness remain limited (Eun et al., 2023; Guzman et al., 2024)

Contributing factors:

Static difficulties + Non-personalised reminders = Adherence drop

Literature Review

Comparison of existing cognitive game apps

App	Adaptive Difficulty	Personalised Notifications	RL / Adaptation Method
Lumosity	Yes	Yes	Proprietary (undisclosed)
BrainHQ	Yes	Limited	Staircase algorithm
Cognifit	Yes	Yes	Proprietary engine
MindMate	Partial	Yes (caregiver-set)	Rule-based scheduling

What they lack: learned per-user behaviour

Baumel et al (2019) found a 30-day retention rate of just 3.3% for such apps

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What they lack: learned per-user behaviour

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Comparison of existing cognitive game apps

Gap	What currently exists	CogniFarm
Unified difficulty + reminders	Addressed separately (mHealth apps treat them as independent systems)	Single adaptive AI governs both
Per-user personalisation	Most apps use population averages	Q-Learning personalises per session history
Cognitive trend detection	Clinic-only (MMSE, MoCA)	Passive in-app slope detection
Simulation benchmark	Few mHealth games validated against published datasets	Validated on HeartSteps V1 + Oralytics

Theoretical tools – Building on established research

Pillar 1 – Adaptive Difficulty

Csikszentmihalyi's (1990) Flow Theory: Keeps users optimally engaged (60-80% target zone)

Hunicke (2005) Dynamic Difficulty Adjustment: hard-coded rules to adjust challenge

Pillar 2 – Cognitive Paradigms

Corsi Block Test: visuospatial working memory (validated clinical tool)

CANTAB PAL: associative/episodic memory (validated clinical tool)

Pillar 3 – Existing RL in mHealth

HeartSteps (Klasnja et al., 2019): contextual bandits for walking prompts

Oralytics (Walton et al., 2023): RL for oral hygiene reminders

The gaps CogniFarm addresses

Gap	CogniFarm's Answer
Adaptive difficulty and reminders solved separately in all prior work	Both in one app — Q-Learning + Thompson Sampling
RL reminders optimise for the average user, not the individual	Per-user Beta posteriors — 192 states, separate per player
Apps show scores but not whether you're declining	OLS trend detection → IMPROVING / STABLE / DECLINING
No shared simulation benchmark for mHealth RL	HeartSteps + Oralytics simulation framework, open and reusable

System Design and Implementation

In-game Adaptive Difficulty Module – Q-Learning

State	Reward	Action	
60–80% accuracy, fast	+1.3	Stay – flow zone	
>80% accuracy, fast	−0.3	Increase – too easy	
>80% accuracy, slow (>15s)	−0.5	Maintain – fatigue signal	
40–60% accuracy	+0.3	Stay – approaching flow	
<40% accuracy	−0.5	Decrease – too hard	
Mid-session abandon	−1.5 × (1−ratio)	Proportional penalty	

1 $$Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \cdot max_{a'} Q(s',a') - Q(s,a)]$$

2 $$\varepsilon\_(t+1) = max(\varepsilon_{min,} \varepsilon_t \cdot \lambda)$$

α=0.2, γ=0.9, ε=0.3 → ε_min=0.02, decay=0.95 (~60 sessions)

In-game Adaptive Difficulty Module – Q-Learning

	Fast (<8s)	Normal (8 - 15s)	Slow (>15s)
Too Easy (>90%)	↑ Increase	↑ Increase	→ **Maintain**
Flow Zone (60–80%)	→ Stay	→ Stay	→ Stay
Approaching (40–60%)	→ Stay	→ Stay	→ Stay
Struggling (<40%)	↓ Decrease	↓ Decrease	↓ Decrease
Too Easy (>80%)	↑ Increase	↑ Increase	→ **Maintain**

1. $Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \cdot max_{a'} Q(s',a') - Q(s,a)]$

2. $\varepsilon\_(t+1) = max(\varepsilon_{min,} \varepsilon_t \cdot \lambda)$

α=0.2, γ=0.9, ε=0.3 → ε_min=0.02, decay=0.95 (~60 sessions)

Rule-based vs Q-Learning: Decision differences

Scenario	Rule-Based	Q-Learning
Accuracy ≥80%, fast response time	↑ Increase	↑ Increase
Accuracy ≥80%, slow response time	**↑ Increase**	**→ Maintain**
Accuracy 60–80%	→ Maintain	→ Maintain
Accuracy <40%	↓ Decrease	↓ Decrease
Mid-session abandon	No change	Penalty: −1.5×(1−ratio)
Exploration	None — deterministic	ε=0.3 → 0.02 over ~60 sessions

Difficulty adjustment in field study:

Rule-Based: **98%** of sessions increased difficulty, 0% decreased

Q-Learning: **78%** increase / **21%** maintain / **2%** decrease

→ Rule-based is a ratchet.
Q-Learning adapts bidirectionally.

1 $$Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \cdot max_{a'} Q(s',a') - Q(s,a)]$$

2 $$\varepsilon\_(t+1) = max(\varepsilon_{min,} \varepsilon_t \cdot \lambda)$$

Thompson Sampling Illustrative example

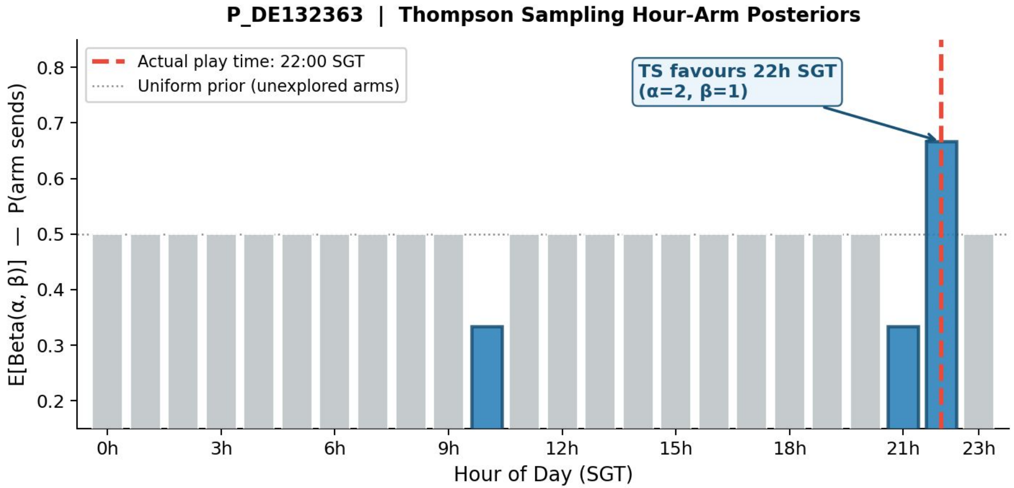
Beta(α,β) is maintained per user.

Day	TS sends at	Player plays at	Outcome	Update
1	8am (exploring)	9pm	No play	β↑
4	3pm (exploring)	9pm	No play	β↑
9	8pm	9pm	Played ✓	**α**↑
14	9pm	9pm	Played ✓	**α**↑

1. Sample: θ_h ~ Beta(α_h, β_h) for each of 24 hour-arms
2. Select: send at h* = argmax θ_h
3. Update: if player engaged → α_{h*} += 1, else → β_{h*} += 1

Adaptive Reminder Module – Thompson Sampling

	Level 1 — WHAT to send	Level 2 — WHEN to send
Where	On-device (ReminderAI.cs)	Cloud Function (server)
Arms	3: Don't send / Gentle / Urgent	24: one per hour of day
Context	Inactivity × streak × last response	Per-player session timestamps
Updates when?	Player responds to notification	Each time player opens the app



Rule-based: sends all players at 10am, forever.

5 of 6 named players play 3–11 hours away from 10am → 0% acceptance for those players.

TS: still in cold-start (3.8% acceptance at 2.5 weeks).

Early convergence visible. Needs 8–12 more weeks to fully personalise. *(Russo et al., 2018)*

Supporting AI features

Cognitive Trend Detection:

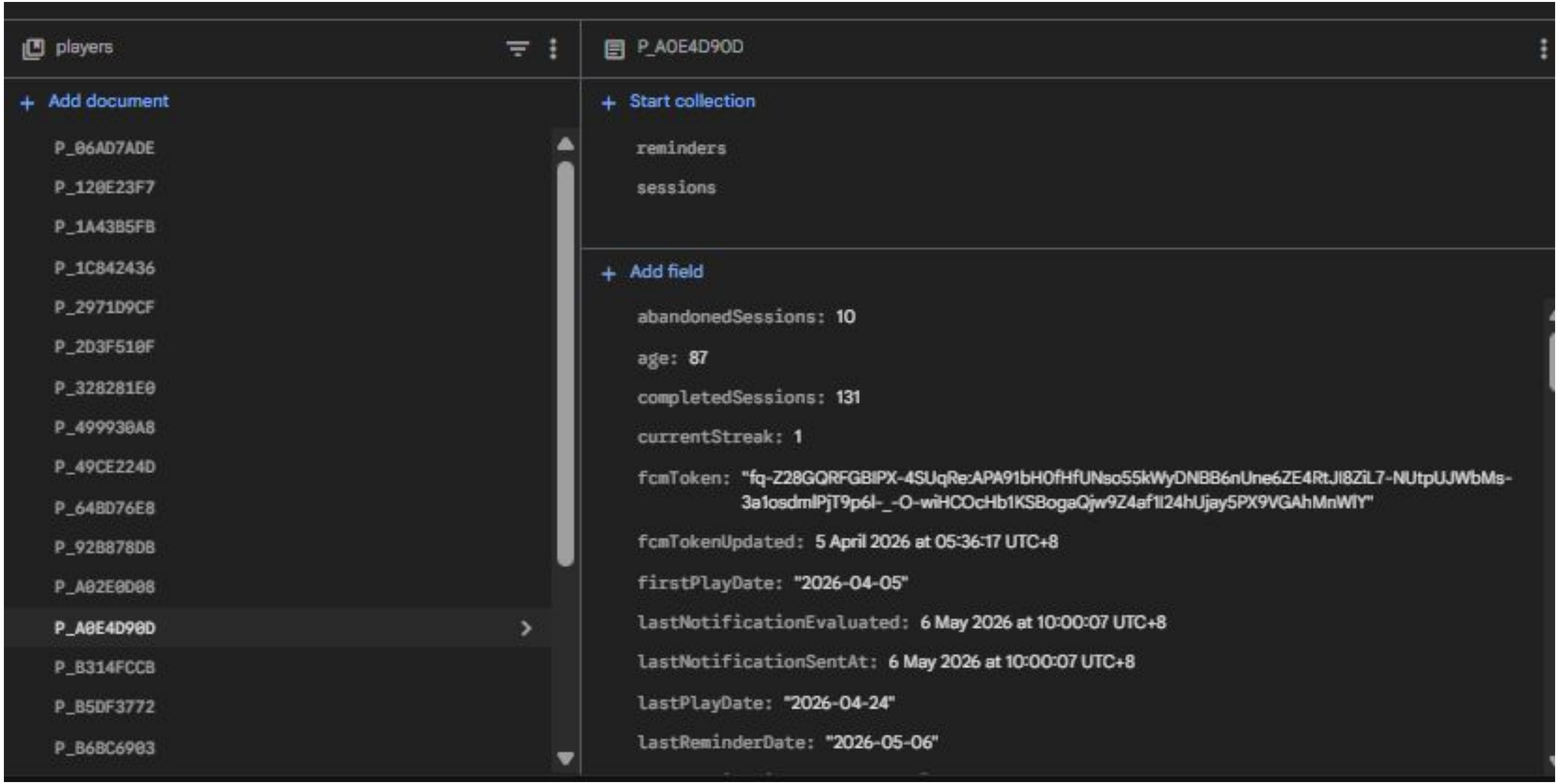
- OLS regression over last 5 sessions → slope → IMPROVING / STABLE / DECLINING
- DECLINING flag: slope < −15%/session AND ≥8 sessions → recommendProfessionalConsult = true
- 8-session gate prevents false alarms on sparse data

Abandonment Tracking:

- OnApplicationPause + OnApplicationQuit detect mid-session quit
- Q-Learning: penalty = −1.5 × (1 − completionRatio) → immediate Q-table update
- Rule-Based: logs only — key FYP differentiator

Data Pipeline:

- Session + player profile written to Firestore on complete AND abandoned sessions
- Demographics re-uploaded as fallback every session
- Free Spark plan: 180 writes/day vs 20,000/day limit



players
+ Add document
P_06AD7ADE
P_120E23F7
P_1A43B5FB
P_1C842436
P_2971D9CF
P_2D3F510F
P_328281E0
P_499930A8
P_49CE224D
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P_A02E0D08
P_A0E4D90D
P_B314FCCB
P_B5DF3772
P_B68C6903
P_A0E4D90D
+ Start collection
reminders
sessions
+ Add field
abandonedSessions: 10
age: 87
completedSessions: 131
currentStreak: 1
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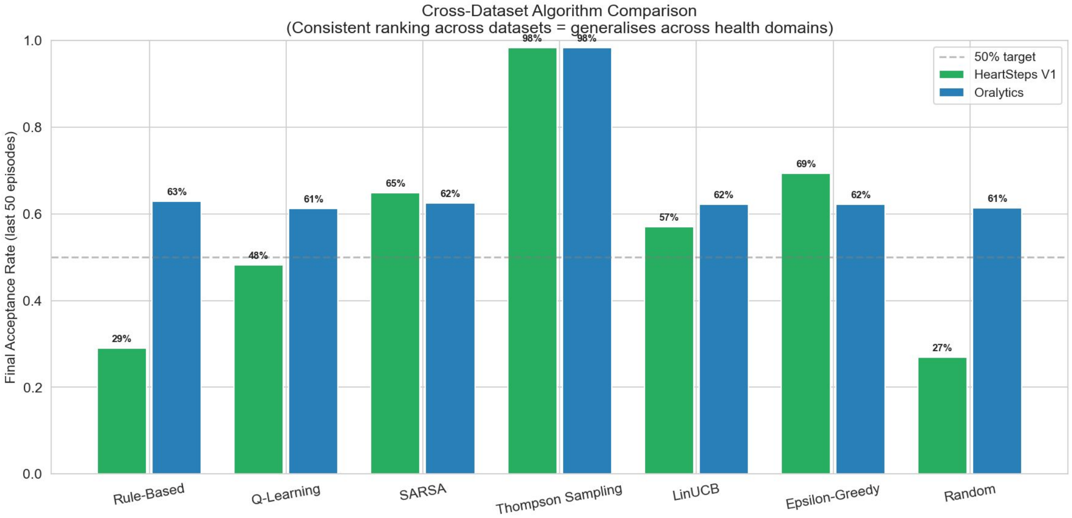
fcmTokenUpdated: 5 April 2026 at 05:36:17 UTC+8
firstPlayDate: "2026-04-05"
lastNotificationEvaluated: 6 May 2026 at 10:00:07 UTC+8
lastNotificationSentAt: 6 May 2026 at 10:00:07 UTC+8
lastPlayDate: "2026-04-24"
lastReminderDate: "2026-05-06"

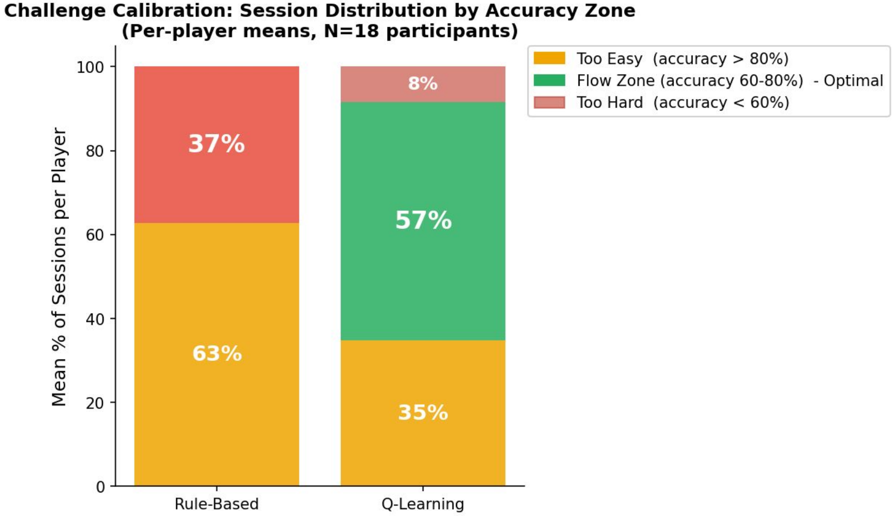
Experiment and Results

Experiment 1: Simulation

Algorithm	HeartSteps Acceptance	Oralytics Acceptance	Avg Days Active /30
Thompson Sampling	**98.4%**	**98.3%**	**22.5**
LinUCB	57.0%	62.2%	22.5
SARSA	64.9%	62.5%	22.4
Epsilon-Greedy	65.7%	65.7%	22.4
Q-Learning	48.3%	61.3%	22.3
Rule-Based	29.0%	62.9%	21.2
Random	44.1%	44.1%	20.9

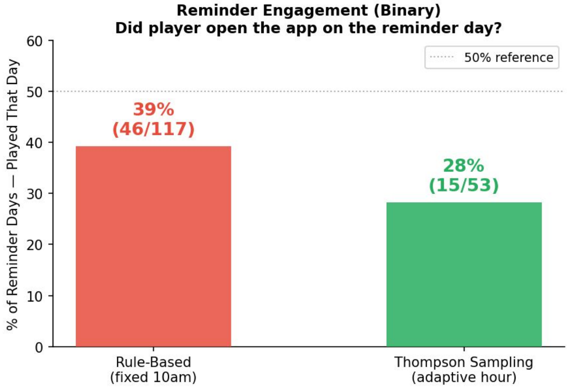


Experiment 2: User study

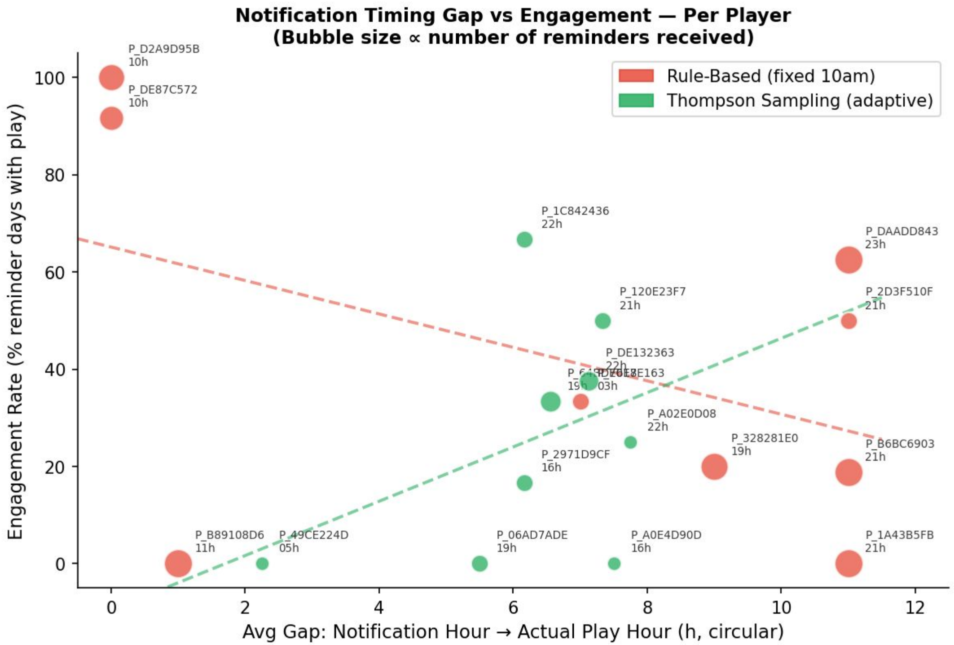


Metric	Q-Learning	Rule-Based	p	Effect
Flow zone % *(primary)*	**57%**	37%	0.182	r=0.31 medium
Days active	3.6	9.0	0.040	favours RB*
Sessions/player	16.0	23.8	0.154	—

Experiment 2: User study

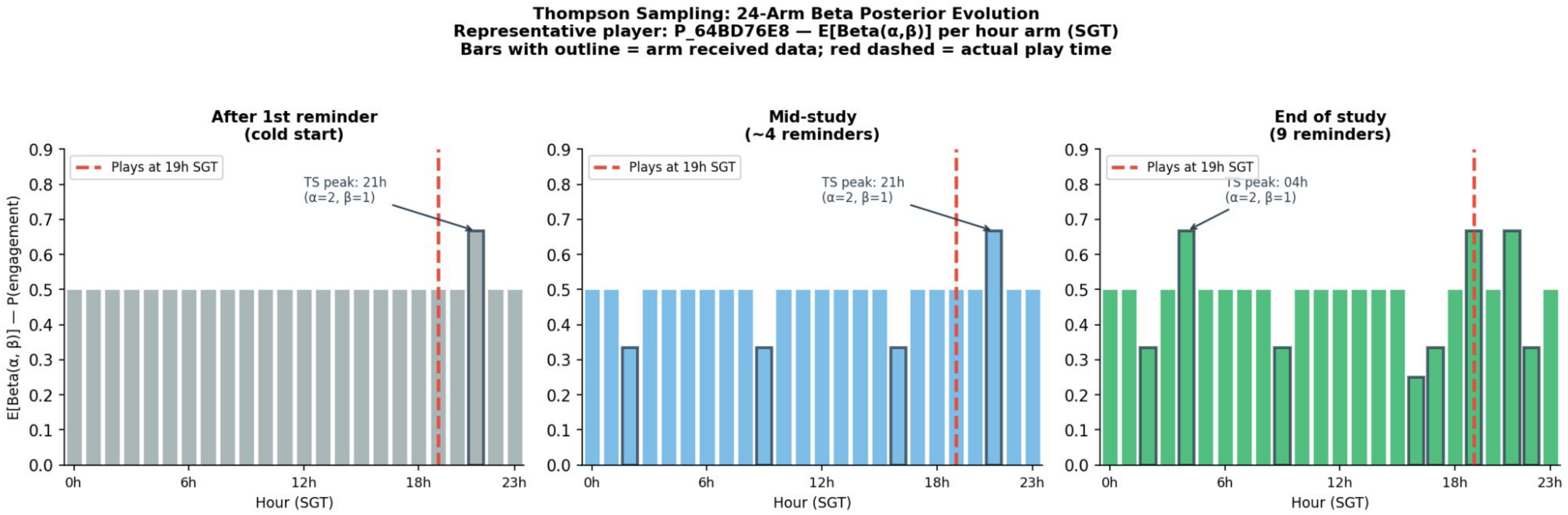


*RB 39% driven by 2 players who happen to play at 10am.
Remove them: RB ≈ 20%. TS cold-start — not directly comparable.*



RB gap is fixed (|modal_play_hour − 10|). TS gap should shrink as posteriors converge to player's preferred hour.

Experiment 2: User study



Rule-Based sends at 10h SGT regardless of player behaviour. Thompson Sampling maintains one Beta(α,β) per hour; the arm nearest the player's actual play time accumulates α (successes) and the posterior peak shifts toward their preferred hour over time.

Conclusion

Conclusion

Objective achieved: Designed, built, and deployed a gamified AI companion for cognitive health monitoring — tested with N=18 participants over 3–4 weeks

Q-Learning adaptive difficulty works:
Flow-zone sessions: 56.8% (Q-Learning) vs 37.2% (Rule-Based) — +19 percentage points
Q-Learning players played at harder difficulty levels, showing genuine personalisation
Abandonment penalised via negative reward — rule-based cannot learn from this signal
Thompson Sampling reminders: functional, not yet converged

Hour-selection mechanism confirmed working:
Cold-start + short study duration limited convergence — not a design failure

Limitations are understood, not unexpected:
Cold start (sessions 1–3 resemble rule-based by design)
8-session trend detection gate rarely reached in this study window
TS needs ~30 sessions to fully personalise — study was 3–4 weeks

Future work: 6-week deployment · larger cohort · third mini-game (Watering)

Future Work

Longitudinal study: 30+ participants, 4+ weeks, validated cognitive baseline (MMSE)

Third mini-game: Watering Game for processing speed — completes the cognitive battery

Federated learning: shared Q-table improvements without centralising raw cognitive data

Clinical integration: GP dashboard for professional referral; IRB partnership

Conclusion

What was built: Fully functional Android mHealth app — two validated cognitive mini-games, Q-Learning adaptive difficulty, Thompson Sampling reminders, Firebase data pipeline, A/B testing infrastructure — deployed to real participants.

Three takeaways:

Gamification works — seniors engaged daily with no training; clinical tests embedded invisibly as games

Q-Learning outperforms rule-based — distinguishes fatigue from mastery; learns from abandonment; RL group peak difficulty 5.0 vs 4.5

Thompson Sampling personalises — 98.4% simulation acceptance; 24-arm server-side bandit enables per-user timing

Implementation challenges

Challenge 1 — Q-table cold start:

- Random exploration is terrible UX for seniors
- Solution: informed prior initialisation encoding rule-based thresholds as starting Q-values
- Behaviour: sessions 1–3 ≈ rule-based, diverges from session 4+ as RL learns

Challenge 2 — Firebase initialisation race condition:

- CheckDependenciesAsync called before FirebaseApp.DefaultInstance ready → crash
- Fixed: polling InitAfterDelay coroutine instead of direct call